Code No: 124CX **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, February - 2024
JAVA PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1.a)	Define polymorphism and inheritance.	[2]
b)	How to cast types in expression evaluation?	[3]
c)	What is an interface and how it differs from a class?	[2]
d)	What is the significance of CLASSPATH variable?	[3]
e)	List out various checked exceptions in Java.	[2]
f)	How to create threads in Java?	[3]
g)	What is the use of Scanner class?	[2]
h)	Contrast text streams and binary streams.	[3]
i)	List the swing containers.	[2]
j)	What are the various events generated by menu, List box components?	[3]

PART-B

(50 Marks)

2. Compare and contrast procedure oriented programming with object oriented programming. Which paradigm is preferred for advanced problem solving? Why. [10]

OR

3.a) What is method overloading? Demonstrate with sample Java code.
b) Write a Java program to check whether a given string is palindrome. [5+5]

4. Write a program to explain the concept of dynamic method dispatch and also discuss real world applications of this concept. [10]

OR

5.a) What is a nested class? Is every nested class an inner class? Justify your answer.
b) Explain the member access rules for package in Java [5+5]

6.a) Explain the benefits of exception handling.
b) Demonstrate creation of user defined exception in Java. [5+5]

OR

7.a) Describe thread states and thread priorities.
b) Write a program to implement producer-consumer communication using inter thread communication functions. [5+5]

8. Write a program to perform find and replace activity on a given file. Read file name as command line argument and string1 to search in the file and replace it with string2. String1 and String2 are from user through prompt. [10]

OR

9.a) Discuss the steps need to be followed in establishing connectivity to a database.

b) What is a hash table? Why is it used in programming? [5+5]

10. What is an event? Explain delegation event model with suitable program for handling mouse events. [10]

OR

11.a) Why is an applet more secure than an application program?

b) Write a program to create user interface for simple arithmetic calculator using swing components and grid layout manager. [4+6]

---ooOoo---